Healthcare Job Market Analysis

Data Cleaning Report

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Date: July 12, 2026
Tool: Google BigQuery (SQL)
Dataset: [LinkedIn Job Postings 2023–2024 (Kaggle)](https://www.kaggle.com/datasets/arshkon/linkedin-job-postings)

1. **Dataset Overview**
   a. Raw dataset row count: 123,842
   b. Rows after cleaning and filtering: 2,854
   c. Original dataset source: https://www.kaggle.com/datasets/arshkon/linkedin-job-postings
   d. BigQuery project: job-market-1.Job_Market.postings
2. **Cleaning Objectives**
   a. Filter the full dataset of 123,842 job postings down to healthcare-relevant roles only, removing non-healthcare postings that appeared due to broad keyword overlap with common job posting language such as the word "healthcare" appearing in employee benefits sections.
   b. Standardize salary data across inconsistent pay period formats. The raw dataset contained a mix of hourly, weekly, biweekly, monthly, and annual compensation values that were not directly comparable without normalization.
   c. Parse and standardize the location column which contained inconsistent formats including city-state, city-country, metropolitan area descriptions, and country-only entries.
   d. Categorize roles into five meaningful role type groups for downstream analysis: Analyst, Manager, Consultant, Coordinator, and Other.
   e. Remove duplicate job postings based on job_id to prevent double-counting in demand and salary analysis.
3. **Issues Found and Resolutions Applied**
   a. Non-Healthcare Roles in Dataset
      i. **Column affected:** description, company_name, title
      ii. **Severity:** High
      iii. **Root cause:** The raw dataset contained job postings across all industries. Simple keyword filtering on terms like "healthcare" and "health" generated false positives because these terms appear in employee benefits sections of non-healthcare job postings (e.g., "competitive healthcare benefits package") rather than in the actual job content.
      iv. **Resolution:** A three-condition filtering approach was implemented:
         1. Pure healthcare companies: A curated list of known healthcare-only organizations including payers (Optum, Humana, Elevance Health, Cigna, Aetna, CVS, Centene, Molina

Healthcare), health systems (HCA Healthcare, CommonSpirit Health, Ascension Health, Kaiser Permanente, Mayo Clinic, Cleveland Clinic), healthcare technology firms (Epic Systems, Cerner, Oracle Health, Cotiviti, Inovalon, Phreesia, Availity), and healthcare consulting firms (Huron, Chartis, Evolent, Mathematica). Postings from these companies are included without requiring description verification since their entire business is healthcare.
    2. Multi-industry firms requiring description verification: Large firms operating across multiple industries including Deloitte, Accenture, EY, KPMG, PwC, McKinsey, Guidehouse, and others were included only when the job description contained specific clinical or operational healthcare terminology — revenue cycle, managed care, population health, prior authorization, utilization management, medical coding, ICD-10, EHR, EMR, Medicaid, Medicare, health system, health plan, payer, value-based care, clinical documentation, risk adjustment, care coordination, health informatics, HIPAA compliance, credentialing, HEDIS, denial management, HCAHPS, and medical records.
    3. Unknown companies with description verification: All companies not appearing on either named list were included only when the job description contained the same specific healthcare operational terminology defined in Condition 2.

  v. **Result:** Raw dataset reduced from 123,842 rows to 2,956 healthcare-relevant postings — a 97.6% reduction reflecting the highly targeted nature of the healthcare operations and analytics focus.

b. Duplicate Job Postings
  i. **Column affected:** job_id
  ii. **Severity:** Medium
  iii. **Root cause:** LinkedIn job posting data commonly contains duplicate entries representing the same role reposted across different time periods or reposted after edits to the original listing. These duplicates would inflate posting volume counts and distort demand analysis if not removed.
  iv. **Resolution:** Deduplication was applied using DISTINCT job_id to identify and retain only the first instance of each unique job_id. Subsequent duplicate entries were excluded from the final dataset.
  v. **Result:** 44 duplicate rows were identified and removed, reducing the final analysis-ready dataset from 2,956 to 2,912 rows.

c. Inconsistent Salary Pay Periods
  i. **Column affected:** max_salary, min_salary, pay_period
  ii. **Severity:** High
  iii. **Root cause:** The raw dataset contained salary values across five different pay period formats — annual, monthly, biweekly, weekly, and hourly — with no standardization. Direct salary comparison or averaging

across these values without normalization would produce meaningless results.

iv. **Resolution:** All salary values were converted to annualized equivalents using the following standardized conversion factors:
    1. HOURLY = × 2,080 (52 weeks × 40 hours)
    2. WEEKLY = × 52 (52 working weeks per year)
    3. BIWEEKLY = × 26 (26 pay periods per year)
    4. MONTHLY = × 12 (12 months per year)
    5. ANNUAL = No conversion, Already annualized
    6. Two new columns were created: max_annual_salary and min_annual_salary, preserving the full salary range for each posting. The original pay_period column was retained for reference and verification.

v. **Known limitation:** Salary outliers may exist if any postings were incorrectly tagged with a pay period that does not match the actual value (e.g., an annual salary incorrectly labeled as hourly would produce an unrealistically high annualized value). Downstream salary analysis queries should apply a reasonable bounds filter — postings with annualized salary outside the $30,000 to $500,000 range should be excluded from average and distribution calculations to prevent distortion from misclassified values.

vi. **Result:** 790 rows have salary data available (salary_available = TRUE), representing 27.1% of the cleaned dataset. The remaining 72.9% of postings did not include salary information. All salary analysis in this project reflects the subset of postings with available compensation data and findings should be interpreted with this limitation in mind.

d. Inconsistent Location Formatting

i. **Column affected:** location

ii. **Severity:** Medium

iii. **Root cause:** The raw location column contained at least four distinct formatting patterns with no standardization:
    1. "City, State" (standard US format, e.g., "Atlanta, GA")
    2. "City, Country" (international format, e.g., "London, United Kingdom")
    3. "X Metropolitan Area" (e.g., "Greater New York Metropolitan Area")
    4. "Greater X" prefix (e.g., "Greater Chicago")
    5. "United States" only (no city or state specified)
    6. These inconsistencies prevented reliable geographic aggregation or state-level analysis without standardization.

iv. **Resolution:** A multi-step cleaning approach was applied using nested REGEXP_REPLACE functions followed by string splitting:
    1. Removed ", United States" suffix from entries formatted as "City, United States"

2. Removed " Metropolitan Area" suffix from entries formatted as "X Metropolitan Area"
3. Removed "Greater " prefix from entries formatted as "Greater X"
4. Entries reading "United States" with no city or state information were set to NULL for the city field
5. Location was split on the comma delimiter to extract city (first element) and state (second element) as separate columns
   Step 6 — Entries where the second element after splitting was a country name rather than a US state abbreviation or state name were set to NULL for state to prevent country names from being treated as state values

v. **Result:** 2,349 rows (80.7% of cleaned dataset) have a usable state value. The remaining 19.3% could not be reliably parsed to a US state due to metropolitan area descriptions, international locations, or entries with insufficient location detail. State-level geographic analysis in this project is based on the 2,349 rows with valid state data.

e. skills_desc Column Excluded

i. **Column affected:** skills_desc

ii. **Severity:** High

iii. **Root cause:** The skills_desc column, intended to contain structured skill tags per posting, was populated in only 2,439 of 123,842 raw rows, a population rate of less than 2%. Analysis built on this column would reflect less than 2% of the dataset and would not be statistically representative of skill demand across the healthcare job market.

iv. **Resolution:** The skills_desc column was excluded entirely from the cleaned dataset. Skill demand analysis for this project is conducted using Python keyword extraction from the description column, which is populated for 100% of raw rows and contains relevant skill and tool mentions embedded in the job description text.

f. Role Categorization

i. **Column affected:** title

ii. **Severity:** Medium

iii. **Root cause:** Job titles in the raw dataset are highly varied with no standardization: "Healthcare Data Analyst," "Sr. Analyst, Clinical Informatics," and "Analyst II, Revenue Cycle" are functionally similar roles but appear as completely distinct values. Direct title-level analysis without grouping would produce hundreds of fragmented categories with insufficient volume for meaningful analysis.

iv. **Resolution:** A CASE statement was applied to the cleaned title field to assign each posting to one of five role categories based on keyword presence in the title:

1. Analyst = 270 (9.4%)
2. Manager = 751 (26.2%)
3. Coordinator = 318 (11.1%)

4. Consultant = 60 (2.1%)
5. Other (director, administrator, specialist, advisor, coder, educator, researcher, planner, liaison, navigator, worker, reviewer, associate) = 1,513 (52.7%)

v. **Note on Other category dominance:** The Other category comprising 52.7% of the dataset reflects the breadth of healthcare role types beyond the four primary categories. Director, specialist, and administrator titles are particularly common in healthcare and account for a significant portion of this group. Downstream analysis will break out the most common titles within Other to provide more granular insight.

g. **Non-Healthcare Roles Passing Through Company-Level Filter**

i. **Column affected:** title, company_name

ii. **Severity:** Medium

iii. **Root cause:** Several companies on the pure healthcare company list (Condition 1) operate across both healthcare and non-healthcare business lines. CVS in particular operates retail pharmacy, retail stores, and healthcare services simultaneously. As a result, retail and non-healthcare job postings from these companies — including titles such as Merchandiser Specialist and PT Sales Associate — passed through the Condition 1 company filter without description verification, since Condition 1 does not require healthcare terminology in the description.

iv. **Resolution:** A title-level exclusion list was added to the WHERE clause in the filtered_postings CTE to remove known non-healthcare retail and service titles that were identified as outliers in the Role Demand by Title analysis. Excluded titles include:

- Merchandiser Specialist
- PT Sales Associate
- Sales Associate
- Merchandiser
- Cashier
- Store Manager
- Pharmacy Technician
- Shift Supervisor

v. **Known limitation:** The exclusion list captures identified outliers but may not be exhaustive. Additional non-healthcare titles from multi-line companies in Condition 1 may remain in the dataset if not identified during exploratory analysis. A more comprehensive long-term fix would add description-level healthcare verification to Condition 1 postings as well, eliminating reliance on the exclusion list.

vi. **Result:** Identified outlier titles removed from dataset prior to role demand and salary analysis. New row count is 2,854

4. Known Limitations

a. **Salary data coverage:** Only 27.1% of cleaned postings include salary information. Salary findings reflect a subset of the healthcare job market and may not be representative of overall compensation patterns. Companies and roles that consistently do not post salaries may be systematically underrepresented in compensation analysis.

b. **Geographic coverage:** 19.3% of postings could not be assigned to a US state due to inconsistent location formatting. State-level geographic analysis reflects 80.7% of the cleaned dataset.
c. **Company name matching:** The three-condition healthcare relevance filter uses exact string matching on company names after applying LOWER(). Variations in company name formatting — abbreviations, subsidiary names, or names with punctuation — may result in some postings from known healthcare companies being excluded (false negatives) or some non-healthcare postings being included (false positives). A spot check of 20 filtered postings per condition confirmed reasonable precision but the filter is not exhaustive.
d. **Skill extraction methodology:** Skill demand analysis relies on Python keyword extraction from unstructured description text rather than a structured skills field. Skill frequency counts reflect keyword presence in descriptions and may overcount skills mentioned in context (e.g., "experience with Epic preferred" versus "Epic required") without distinguishing between required and preferred qualifications.
e. **Dataset recency:** The LinkedIn Job Postings dataset covers 2023–2024. Healthcare hiring conditions, role demand, and compensation benchmarks may have shifted since the data was collected. Findings should be interpreted as representative of the 2023–2024 healthcare job market rather than current conditions.

5. **Final Dataset Summary**
   a. Raw rows: 123,842
   b. Final analysis-ready rows: 2,854
   c. Rows with salary data: 790 (27.7%)
   d. Rows with valid state: 2,349 (82.3%)
   e. Analyst roles: 270 (9.5%)
   f. Manager roles: 751 (26.3%)
   g. Coordinator roles: 318 (11.1%)
   h. Consultant roles: 60 (2.1%)
   i. Other roles: 1,513 (53%)
6. **Next Steps**
   a. With cleaning complete the following analysis phase tasks are ready to execute:
      i. BigQuery SQL analysis queries answering three SMART analytical questions: role demand by title and geography, skill frequency from description text, and salary variation by role type
      ii. Python keyword extraction in Google Colab using the cleaned description column to identify the most in-demand skills across healthcare operations and analytics postings
      iii. Power BI dashboard construction connecting to cleaned BigQuery output